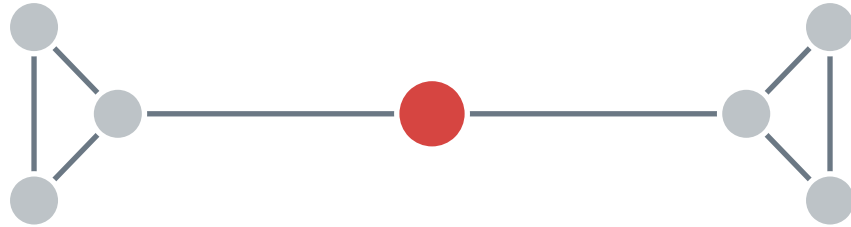


Three-Level Centrality Framework (Adapted from Jang & Yang 2023)

MACRO: Betweenness Centrality

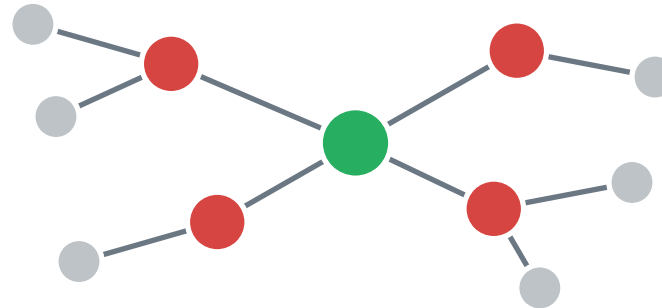


"Which states bridge regional clusters?"

High betweenness node connects otherwise disconnected regions

Example: Texas, Pennsylvania, Washington

MESO: Eigenvector Centrality

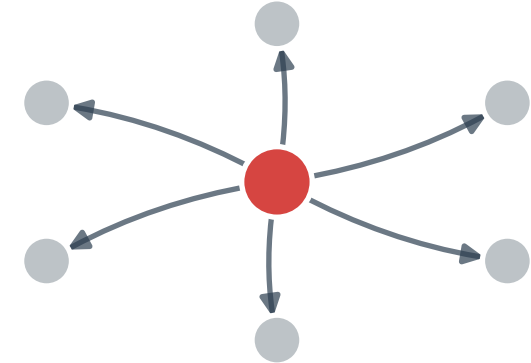


"Which states trade with powerhouses?"

High eigenvector node connected to other important nodes

Example: Texas, Ohio, Florida

MICRO: Weighted Out-Degree



"Which states have highest production capacity?"

High out-degree node distributes large volumes to many destinations

Example: California, Illinois, Indiana

■ High centrality ■ High eigenvector ■ Low/baseline